

Problem

An 500-kW induction furnace at 0.88 power factor operates 20 hours per day for 26 days in a month. Determine the electricity bill per month based on the tariff in Example 11.18.

Example 11.18:

A 300-kW load supplied at 13 kV (rms) operates 520 hours a month at 80 percent power factor. Calculate the average cost per month based on this simplified tariff:

Energy charge: 6 cents per kWh

Power-factor penalty: 0.1 percent of energy charge for every 0.01 that pf falls below 0.85.

Power-factor credit: 0.1 percent of energy charge for every 0.01 that pf exceeds 0.85.

Solution:

The energy consumed is

$$W = 300 \text{ kW} \times 520 \text{ h} = 156,000 \text{ kWh}$$

The operating power factor $\text{pf} = 80\% = 0.8$ is 5×0.01 below the prescribed power factor of 0.85. Since there is 0.1 percent energy charge for every 0.01, there is a power-factor penalty charge of 0.5 percent. This amounts to an energy charge of

$$\Delta W = 156,000 \times \frac{5 \times 0.1}{100} = 780 \text{ kWh}$$

The total energy is

$$W_t = W + \Delta W = 156,000 + 780 = 156,780 \text{ kWh}$$

The cost per month is given by

$$\text{Cost} = 6 \text{ cents} \times W_t = \$0.06 \times 156,780 = \$9,406.80$$

Step-by-step solution

Step 1 of 4

Calculate the value of the energy consumed W .

$$\begin{aligned} W &= 500 \text{ kW} \times 26 \text{ days} \times 20 \text{ hours} \\ &= 260000 \text{ kWh} \end{aligned}$$

Therefore, the value of the energy consumed W is 260000 kWh.

Step 2 of 4

The operating power factor 0.88 is greater than 0.85 by 0.03. Hence, the credit factor is 0.3%.

Calculate the energy charge.

$$\begin{aligned} \Delta W &= 260000 \times \frac{0.3}{100} \\ &= 780 \text{ kWh} \end{aligned}$$

Therefore, the value of the energy charge ΔW is 780 kWh.

Step 3 of 4

Calculate the value of total energy charge.

$$W_t = W - \Delta W$$

Subtraction is used for the power factor credit.

Substitute 780 kWh for ΔW and 260000 kWh for W .

$$\begin{aligned} W_t &= 260000 \text{ k} - 780 \text{ k} \\ &= 259220 \text{ kWh} \end{aligned}$$

Therefore, the value of the total energy charge W_t is 259220 kWh.

Step 4 of 4

Calculate the amount of electricity bill per month.

$$\begin{aligned} C &= 259220 \text{ k} \times 0.06 \\ &= \$15553.20 \end{aligned}$$

Therefore, the amount of electricity bill per month C is \$15553.20.